Listwise Playlist Reordering via Birkhoff-Polytope Relaxation over a Frozen Skip-Probability Model

Divek Patel Sunny Wang Carol Tu Peter Shen

May 6, 2026

Abstract

Music streaming platforms use ordering as a product lever: the same catalogue of tracks produces very different listener experiences depending on the sequence in which they are played. We cast listwise playlist reordering as a combinatorial optimization problem over the $n!$ permutations of a candidate set, where the objective is to minimize the cumulative predicted skip probability under a learned scorer. We approach this problem with two separate tasks. First, we train a transformer-based skip-probability model on the Spotify Sequential Skip Prediction dataset [3], iterating through four variants to arrive at a history-encoder plus current-event-context fusion model with a no-leakage feature set, which attains a skip F_1 of 0.8009. Second, we freeze that scorer and relax the feasible set from permutation matrices to the Birkhoff polytope of doubly-stochastic matrices [2, 12]. A free logit matrix is projected to the polytope via log-space Sinkhorn normalization [4, 18], optimized with Adam [10], and rounded with the Hungarian algorithm [8, 11] together with 32 Gumbel-Sinkhorn samples [15]. On 2,000 held-out sessions of length 10–20, the Birkhoff pipeline reduces mean expected skips from 7.52 (identity) to 5.12, a 32.98% relative reduction. Additionally, it improves on the original order in 97.7% of sessions and on a quadratic greedy baseline in 93.0%. The two-module design keeps the gradient signal semantically clean, keeps the evaluation metric honest, and scales to longer playlists than enumeration or greedy search.

1 Introduction

Skipping a track is one of the clearest negative signals a streaming listener emits. Accumulated skips across a session correlate with shorter engagement, reduced retention, and lower satisfaction, and they shape downstream objectives from recommendation reranking to inventory-level content acquisition. Ordering decisions for Spotify in products such as Daily Mix, Radio, and Discover Weekly are therefore a direct product lever: rearranging the same set of tracks can meaningfully change a session's outcome without touching the catalogue.

Thus, we study the following concrete question: given a fixed session of n candidate tracks for a particular listener, together with per-track features and contextual signals, which permutation of the n tracks minimizes the listener's expected number of skips under a learned skip-probability model? Even for a small subset of $n = 20$, naive enumeration of $n! \approx 2.4 \times 10^{18}$ permutations is infeasible. A greedy position-by-position search reduces this to $O(n^2)$ scorer calls, but it commits to local decisions early in the sequence that constrain later ones, and its improvements over identity are small in practice.

Our approach decomposes the problem into two independent modules:

Skip-Probability Scorer. We train a causal transformer [19] on the Spotify Sequential Skip Prediction dataset [3] through a principled ablation sequence. Starting from a compound-token model with joint next-song and skip prediction, we iterate to a direct per-track skip objective, then add a fusion branch that conditions on the current song and the current-event context in addition to prior session history. A permutation-importance audit reveals that one categorical feature, `hist_user_behavior_reason_end`, leaks the skip label; we drop it from both the song-level auxiliary profile and the event-context input. The resulting no-leak model, which we denote f, achieves a skip F_1 of 0.8009 on held-out sessions.

Birkhoff-Polytope Ordering Optimizer. We freeze f and pose the reordering problem as a continuous optimization over the Birkhoff polytope $\mathbb{B}_n$—the convex hull of $n \times n$ permutation matrices [2]. We parameterize

a doubly-stochastic matrix $\mathbf{P}$ by a free logit matrix $\mathbf{L} \in \mathbb{R}^{n \times n}$ projected onto $\mathbb{B}_n$ via log-space Sinkhorn iterations [4, 18]. A differentiable soft forward through the frozen scorer yields a gradient signal that we feed to Adam algorithm [10]. We round the converged $\mathbf{P}$ to an integer permutation with the Hungarian / Jonker-Volgenant algorithm [8, 11] and complement it with 32 Gumbel-Sinkhorn samples [15] that probe alternative modes of $\mathbf{P}$. Every reported number comes from a hard, integer-indexed forward pass through f. Note, the soft relaxation is only a gradient source.

Our contributions are:

- A four-variant ablation arriving at a context-fused transformer skip scorer with an explicit no-leakage audit, reaching skip $F_1 = 0.8009$ on the WSDM Cup 2018 sessions.
- A complete Birkhoff-polytope reordering pipeline, including log-space Sinkhorn projection, Adam on free logits with NaN-guarding, and parallel Hungarian plus Gumbel-Sinkhorn rounding.
- An empirical study on 2,000 held-out sessions showing the Birkhoff pipeline reduces expected skips from 7.52 to 5.12 (32.98% relative), beating identity on 97.7% and a quadratic greedy baseline on 93.0% of sessions.
- A concrete articulation of why the scorer must be treated as a frozen black box, covering identifiability, gradient semantics, environment–policy separation, and evaluation honesty.

2 Related Work

Sequential recommendation has been studied extensively with recurrent [6] and self-attentive [9] architectures that model the session as an ordered token stream. The Spotify Sequential Skip Prediction Challenge [3] specifically targets per-track skip prediction and released the session logs we use. Most prior work on this data focuses on predicting skips under the realized ordering. Instead, we take the predictor as a given and ask how to choose the ordering itself.

Several prior works relax permutations to doubly-stochastic matrices. Sinkhorn propagation [1] and DeepPermNet [16] use iterative Sinkhorn normalization to make ranking differentiable. Gumbel-Sinkhorn networks [15] combine the Gumbel-Softmax reparameterization [7, 14] with Sinkhorn normalization to sample and optimize over permutations. NeuralSort [5] provides a different continuous relaxation of sorting. Our optimizer is closest in spirit to Gumbel-Sinkhorn but with two domain-specific modifications: we optimize at inference time for a single black-box objective (not end-to-end) and we use the Gumbel machinery only for rounding, not for training.

The Birkhoff–von Neumann theorem [2] identifies the doubly-stochastic matrices as the convex hull of permutation matrices, which makes $\mathbb{B}_n$ a natural home for permutation-valued optimization. Lim and Wright [12] give extension-complexity results that motivate using $\mathbb{B}_n$ as the feasible set even when the objective is non-convex. Modern ML-style Sinkhorn computation traces back to Cuturi [4], who popularized entropic regularization for optimal transport.

3 Problem Formulation

A session is an ordered tuple of n tracks with per-track features $x_1, \ldots, x_n$ (song identity, normalized numeric attributes, and categorical attributes) together with per-track event-context features $c_1, \ldots, c_n$ describing how each track was reached. A learned scorer f takes the full session under some ordering and returns $f(\sigma, i) \in [0, 1]$, the probability that the track placed at position i is skipped, for $i = 1, \ldots, n-1$. Position 0 carries no history and is excluded from the objective. Let $\mathcal{S}_n$ denote the set of permutations of $\{1, \ldots, n\}$.

Discrete Objective. We seek the permutation that minimizes cumulative predicted skip:

$$\sigma^\star \;=\; \arg\min_{\sigma \in \mathcal{S}_n} \sum_{i=1}^{n-1} f(\sigma, i). \tag{1}$$

The search space $|\mathcal{S}_n| = n!$ is intractable even for moderate n; explicit enumeration becomes prohibitive beyond $n \approx 12$.

Continuous Relaxation. Encode each σ as an $n \times n$ permutation matrix M_σ with $M_\sigma[i,j] = 1$ iff track j is placed at position i. The Birkhoff polytope

$$\mathbb{B}_n \;=\; \left\{ \mathbf{P} \in \mathbb{R}_{\geq 0}^{n \times n} \;\middle|\; \mathbf{P}\mathbf{1} = \mathbf{1},\; \mathbf{1}^\top \mathbf{P} = \mathbf{1}^\top \right\} \tag{2}$$

is the convex hull of all such M_σ [2]. We relax $M_\sigma \to \mathbf{P} \in \mathbb{B}_n$, define a differentiable soft extension $\tilde{f}(\mathbf{P})$ of $\sum_i f(\sigma, i)$, minimize $\tilde{f}$ over $\mathbb{B}_n$ with gradient descent, and round the solution back to $\mathcal{S}_n$. The final metric is always Eq. 1 evaluated at the rounded permutation. Note, the relaxation is a gradient source, not an evaluation substitute.

4 Data and Features

We use the Music Streaming Sessions Dataset released for the WSDM Cup 2018 Sequential Skip Prediction Challenge [3], joined with the accompanying track metadata and audio features. After merging session logs with track records we retain 167,880 event rows, which we partition session-wise into 7,000 training, 1,000 validation, and 2,000 test sessions. The session-level split uses `GroupShuffleSplit` on the session identifier so that no session appears in more than one split.

We group features into three families capturing *where* the listener is in the session, *how smooth* the musical transitions are, and *what* the listener has been doing. The first family includes session progress, hour of day, day of week, a weekend flag, and session length. The second comprises per-track audio features (acoustic-vector components, beat strength, dynamic range, valence, liveness, organism, bounciness, duration) that jointly capture transition smoothness. The third combines cumulative skips and skip rate so far, pause-before-play flags (`short_pause`, `long_pause`, `no_pause`), shuffle and premium indicators, seekback count, and context type.

Numeric features are normalized to training-set mean and standard deviation. Categorical features use train-only vocabularies with `<UNK>` fall-throughs. Any song that appears fewer than a minimum number of times in training is mapped to an `<UNK_SONG>` token.

A permutation-importance analysis against the initial context-fused model (Section 5) revealed that the categorical feature `hist_user_behavior_reason_end` dominated all other features by an order of magnitude in its contribution to skip F_1. This variable encodes *how* the current track's playback ended, which is essentially the skip label. We drop it from both the song-level auxiliary profile and the event-context categorical inputs. Every experiment in Section 8 uses this no-leak feature set, and all reorderings are evaluated with the corresponding no-leak scorer f.

5 Skip-Probability Scorer

We describe the scorer through four progressively stronger variants. Each variant shares the same causal transformer backbone, but what changes is the training target, the input representation, and the output head.

All four variants use a 6-layer pre-norm causal transformer with model dimension $d_{\text{model}} = 256$, eight attention heads, feedforward dimension 1024, maximum sequence length 40, and no bias terms. We use RMSNorm [20] on the residual branches and a SwiGLU [17] feedforward inside each block. Optimization uses AdamW [13].

Variant A: Next-token Compound Model. The first variant follows a classic next-token framing: given a causally-masked history of song and skip tokens, predict whether the next event is the skip token or a specific song token. Song-token embeddings are compound, $\mathrm{E}_{\text{song}}(\text{title}) + \mathrm{P}_{\text{num}}(\text{aux num}) + \sum_j \mathrm{E}_{\text{cat}}(\text{aux cat}_j) + \mathrm{E}_{\text{pos}}$, and the skip token has its own learned embedding. Two output heads share the backbone: a song softmax and a binary skip head. The loss is a convex combination of the two cross-entropies, with the skip term up-weighted by inverse class frequency. At test time we decode via a threshold on $p(\text{skip})$. This variant reaches a skip F_1 of 0.6878.

Variant B: Direct Per-track Skip Objective. Variant A optimizes a proxy: the probability the *next token* is the skip token, which couples skip prediction to song-identity prediction. Variant B decouples them by redefining each example as a (prior history, current track) pair with the binary skip label of the current track

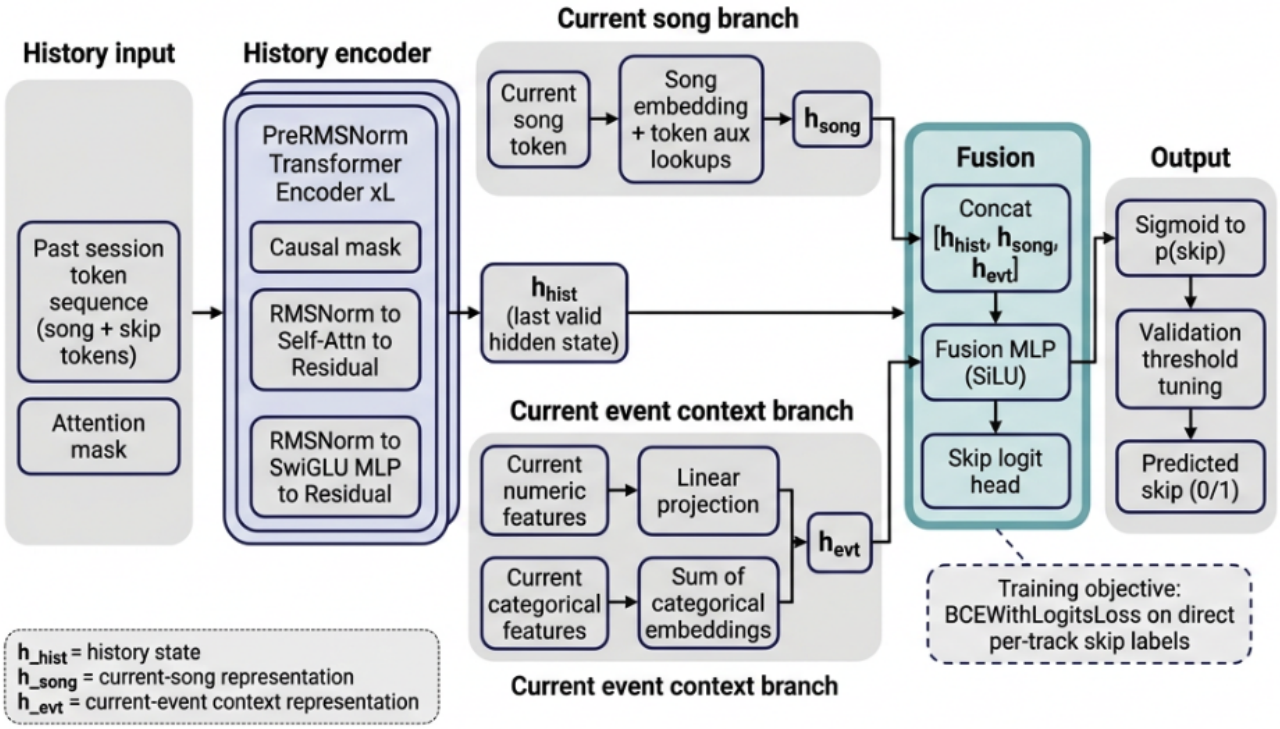

Figure 1: Architecture of the context-fused skip scorer (Variant C). The history encoder produces h_{hist} from the causally-masked prior-event sequence. The fusion block concatenates all three input representations, passes them through an MLP with SiLU activation, and emits a single skip logit trained with `BCEWithLogitsLoss` on direct per-track skip labels.

as target. The loss becomes a pure `BCEWithLogitsLoss` with `pos_weight` offsetting class imbalance. Skip F_1 rises to 0.7066.

Variant C: History and Current-event Context Fusion. The biggest jump in performance comes from enriching the input representation. In addition to the causal history encoder, we add two further representations of the *current* event: an embedding of the current song token (through the shared song-token machinery), and a projection of the current numeric and categorical event-context features. These three representations are concatenated and fused through a linear layer with a SiLU nonlinearity, followed by a binary skip head. Formally,

$$p_{\text{skip}} \;=\; \sigma\big(W_{\text{out}} \cdot \text{SiLU}(W_{\text{fuse}}[h_{\text{hist}};\, h_{\text{song}};\, h_{\text{evt}}])\big). \tag{3}$$

Figure 1 shows the fused architecture. Skip F_1 jumps to 0.8763 on the leaky feature set.

The leakage audit described in Section 4 identifies `hist_user_behavior_reason_end` as near-label information. Dropping it and retraining the Variant C architecture yields a skip F_1 of 0.8009 at a tuned threshold of 0.43. A permutation-importance analysis of the resulting model confirms that the remaining dominant features (`reason_start`, pause-before-play flags, seekback count) are behavioral rather than outcome-adjacent, consistent with the goal of predicting skips without observing their proximate cause. We denote this model f throughout the paper; it is the frozen black box used for all Birkhoff optimization and evaluation.

Table 1 summarizes the ablation. Precision and recall are computed at each variant's own tuned threshold.

6 Birkhoff-Polytope Playlist Reordering

Figure 2 provides the overall playlist reordering pipeline. Four stages operate in sequence: feature preparation (gray), Adam optimization on the Birkhoff polytope (turquoise), rounding from the polytope to a pool of

Table 1: Skip-scorer ablation on held-out sessions. Precision, recall, and F_1 are skip-class metrics; threshold is tuned on validation.

Variant	Thr.	Prec.	Rec.	F_1
A. Next-token compound	0.60	0.689	0.686	0.688
B. Direct per-track skip	0.45	0.694	0.720	0.707
C. Context fusion (leaky)	0.51	0.849	0.906	0.876
C′. Context fusion (no-leak)	0.43	0.752	0.857	**0.801**

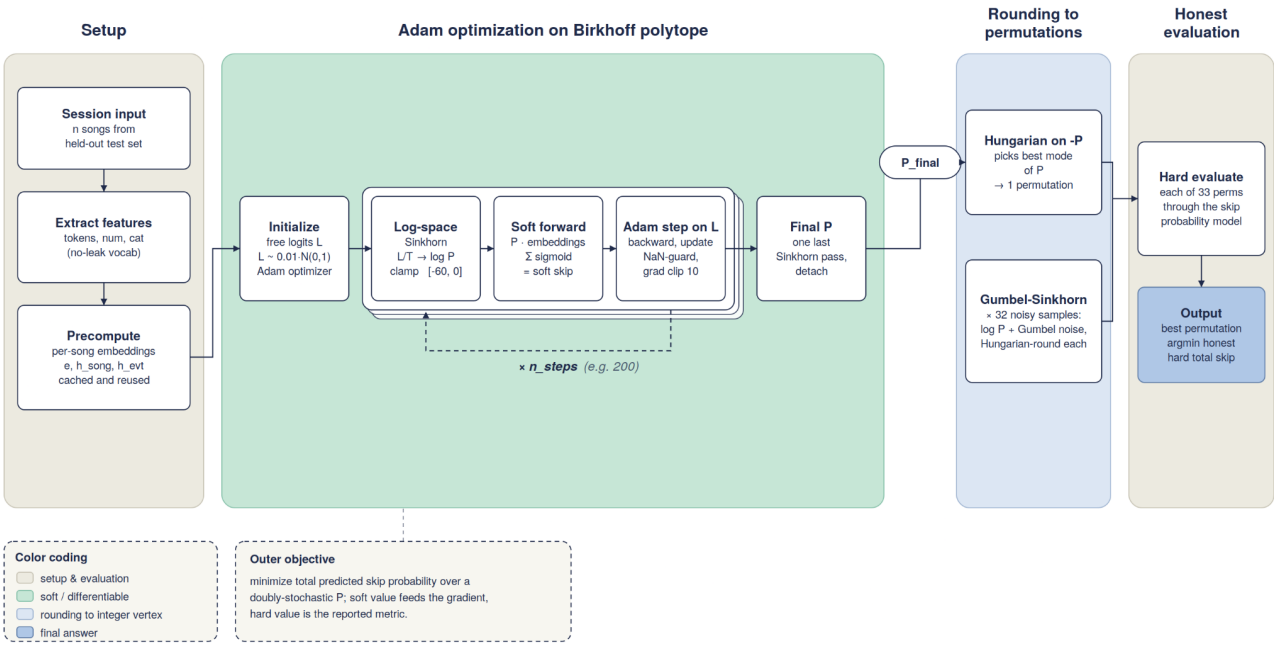

Figure 2: Pipeline for Birkhoff-Polytope Playlist Reordering. A session of n held-out tracks is featurized and its per-song embeddings are cached. A free logit matrix $\mathbf{L}$ is projected to the polytope by log-space Sinkhorn normalization, fed through a soft forward pass of the frozen scorer to produce a gradient signal, and updated by Adam. After n_{steps} iterations, the final doubly-stochastic $\mathbf{P}$ is rounded to integer permutations using the Hungarian algorithm on $-\mathbf{P}$ (one candidate) and 32 Gumbel-Sinkhorn samples on $\log \mathbf{P}$. All 33 candidates are evaluated by a hard, integer-indexed forward pass through the frozen scorer; the minimum is reported.

integer permutation matrices (light blue), and final honest hard evaluation (beige). Only the optimization stage differentiates through f and every metric we report comes from the honest evaluation stage.

6.1 Freezing the Scorer

Treating the scorer as a fixed black box is a modeling choice that provides three main benefits.

Identifiability. If the scorer's weights could co-adapt during Adam on $\mathbf{L}$, the trivial minimizer of "expected skip probability" is a scorer that outputs zero everywhere. The loss would vanish while saying nothing about whether the ordering is actually better. Freezing f ensures the only way to reduce the objective is to find orderings that are genuinely better under a fixed definition of better.

Clean gradient semantics with an environment–policy split. Gradients of $\tilde{f}(\mathbf{P})$ flow through f into $\mathbf{L}$ but never update f. Therefore, the gradient has a single clean meaning: how should $\mathbf{P}$ change so that this fixed listener model predicts fewer skips? The setup mirrors offline reinforcement learning with a fixed simulator as the scorer is the environment and the permutation is the action, and we do not retrain the environment while searching for good actions inside it. If f were trainable, gradients would mix the policy signal with "how should

my beliefs about skipping change?"

Evaluation honesty. Before-and-after skip rates are comparable only when they come from the same f. A single frozen scorer makes the Birkhoff number and the identity number refer to the same base measurement device.

6.2 Parameterization and Log-Space Sinkhorn

We parameterize $\mathbf{P}$ as a free logit matrix $\mathbf{L} \in \mathbb{R}^{n \times n}$ scaled by a temperature $T > 0$ and projected onto $\mathbb{B}_n$ by iterative Sinkhorn normalization [18]. Sinkhorn's theorem guarantees that alternating row and column normalization of a positive matrix converges to a doubly-stochastic matrix. We perform the iterations in log space:

$$\log \mathbf{P} \leftarrow \log \mathbf{P} - \mathrm{logsumexp}_{\mathrm{col}}\big(\log \mathbf{P}\big), \tag{4}$$

$$\log \mathbf{P} \leftarrow \log \mathbf{P} - \mathrm{logsumexp}_{\mathrm{row}}\big(\log \mathbf{P}\big), \tag{5}$$

starting from $\log \mathbf{P} = \mathbf{L}/T$, repeated for a fixed number of inner iterations and bounded to $[-60, 0]$ so that $\exp(\log \mathbf{P})$ cannot underflow beyond $\approx 10^{-26}$ while still being effectively zero for a permutation.

We found this log-space formulation essential as a naive multiplicative Sinkhorn underflows at the low temperatures we need to sharpen $\mathbf{P}$ toward a vertex, and the resulting $\mathbf{P}$ cannot be fed to a downstream $\log \mathbf{P}$ computation without producing $-\infty$ entries that propagate as NaNs. Working in log space with `logsumexp` avoids both potential failures.

6.3 Soft Forward Pass

Given a doubly-stochastic $\mathbf{P}$, we define the soft forward $\tilde{f}(\mathbf{P})$ by using $\mathbf{P}$ to form convex mixtures of per-song representations. Concretely, for each position i we compute a soft token embedding $\tilde{h}_{\mathrm{song}}[i] = \sum_j \mathbf{P}[i, j]\, h_{\mathrm{song}}[j]$ and a soft event-context embedding $\tilde{h}_{\mathrm{evt}}[i]$ in the same way. The history encoder is applied to the resulting soft sequence, the three branches are fused, and a sigmoid produces a per-position soft skip probability where the soft total is their sum.

This is an *off-distribution approximation*. The scorer was trained on integer tokens, not convex mixtures, so $\tilde{f}$ is not calibrated as a probability and systematically underestimates the hard total. We use $\tilde{f}$ only to produce a gradient signal for $\mathbf{L}$. All reported metrics use hard (integer-indexed) permutations.

6.4 Optimization

We optimize $\mathbf{L}$ with Adam [10], initialized as $\mathbf{L} \sim 0.01 \cdot \mathcal{N}(0, 1)$, for $n_{\mathrm{steps}} = 200$ iterations at learning rate 0.05, with 30 Sinkhorn inner iterations per step and temperature $T = 0.7$. In each step, we run the log-space Sinkhorn on $\mathbf{L}/T$, take the soft forward, backpropagate into $\mathbf{L}$, and step Adam. We sanitize $\mathbf{L}$ and its gradient with `torch.nan_to_num` and clip gradients to norm 10, both of which prevent a single numerical blow-up from corrupting Adam's running moments.

6.5 Rounding to Permutation Matrices

The converged $\mathbf{P}$ is close to, but rarely a vertex. We round it in two ways and merge the results.

Hungarian. The Jonker–Volgenant variant [8] of the Hungarian algorithm [11] on the cost matrix $-\mathbf{P}$ returns the permutation σ that maximizes $\sum_i \mathbf{P}[i, \sigma(i)]$. This gives us the single vertex closest to the converged doubly-stochastic solution $\mathbf{P}$.

Gumbel-Sinkhorn Sampling. Hungarian on $-\mathbf{P}$ picks one mode, but the soft optimum often lies on several nearby modes. We draw 32 additional candidates with Gumbel-Sinkhorn sampling [15]: add i.i.d. Gumbel noise $G_{ij} = -\log(-\log U_{ij})$ with $U_{ij} \sim \mathrm{Uniform}(0, 1)$ to $\log \mathbf{P}$ and round each noisy copy with Hungarian. The noise scale is 1.0.

This yields 33 candidate permutation matrices in total for final evaluation.

Table 2: Main results over 2,000 held-out sessions of length 10–20. Totals are sums of per-position skip probabilities under the frozen scorer f; 95% CI is for the mean across sessions. Lower is better. Runtime is mean wall-clock per session.

Method	Mean skips (95% CI)	% Reduction	Time/sess
Identity	7.5181 [7.358, 7.678]	0.00%	< 0.01 s
Greedy	6.9298 [6.774, 7.086]	7.55%	≈ 1.0 s
Birkhoff	**5.1244 [4.993, 5.256]**	**32.98%**	≈ 6.0 s

6.6 Honest Evaluation and Final Selection

Each of the 33 candidates is scored by running the *frozen* scorer f with integer token indices—no mixture, no soft forward. We report the minimum total skip probability across candidates as the Birkhoff result for that session. This is the honest number; it is always $\geq \tilde{f}(\mathbf{P})$ because the soft forward biases downward.

7 Experimental Setup

We evaluate on the 2,000 held-out test sessions described in Section 4, filtered to session lengths in $[10, 20]$. This range is small enough that the $O(n^2)$ greedy baseline remains tractable and large enough that enumeration is infeasible (20! permutations for the longest sessions).

We compare against two baselines that share the frozen scorer f:

Identity: Evaluates the original session ordering. This is the natural null baseline; the observed skip rate in the raw data is a function of whatever ordering the platform actually served.

Greedy: Fixes the first position (which carries no history and therefore does not contribute to the objective), then for $i = 1, \ldots, n-1$ picks the remaining track that minimizes the scorer's skip probability at position i given the already-placed prefix. This is $O(n^2)$ scorer calls per session.

Both baselines call the same f used by the Birkhoff optimizer; differences in the reported totals reflect only the search strategy.

Birkhoff hyperparameters were selected on a three-session tuning subset by grid search over six configurations spanning different step counts, learning rates, Sinkhorn iteration counts, temperatures, and entropy regularization strengths. The selected configuration is $n_{\text{steps}} = 200$, $\text{lr} = 0.05$, $\text{sinkhorn_iters} = 30$, $T = 0.7$, no entropy regularization. Rounding uses 32 Gumbel-Sinkhorn samples with noise scale 1.0, giving 33 candidates.

The session split uses random state 42; hyperparameter-tuning session selection uses random state 12,345; Birkhoff initialization uses seed 0; Gumbel-Sinkhorn sampling uses seed 1.

8 Results

Table 2 gives the main comparison. Across 2,000 test sessions, identity yields 7.52 expected skips per session on average; greedy reduces this to 6.93 (7.55% relative reduction); Birkhoff reaches 5.12, a 32.98% relative reduction. The 95% confidence intervals are non-overlapping between all three methods.

Birkhoff achieves a lower expected skip total than identity on 1,954/2,000 sessions (97.7%), and matches or beats greedy on 1,860/2,000 sessions (93.0%). In a 20-track session the mean reduction relative to identity is approximately 2.4 expected skips.

Figure 3 shows the full distribution of per-session skip totals. The left panel is a histogram; the Birkhoff distribution is shifted left and more concentrated than identity or greedy. The middle panel shows the cumulative distributions; the Birkhoff curve lies uniformly above the others at every threshold, which is a much stronger statement than a difference in means. The right panel gives box plots: Birkhoff's median, quartiles, and mean

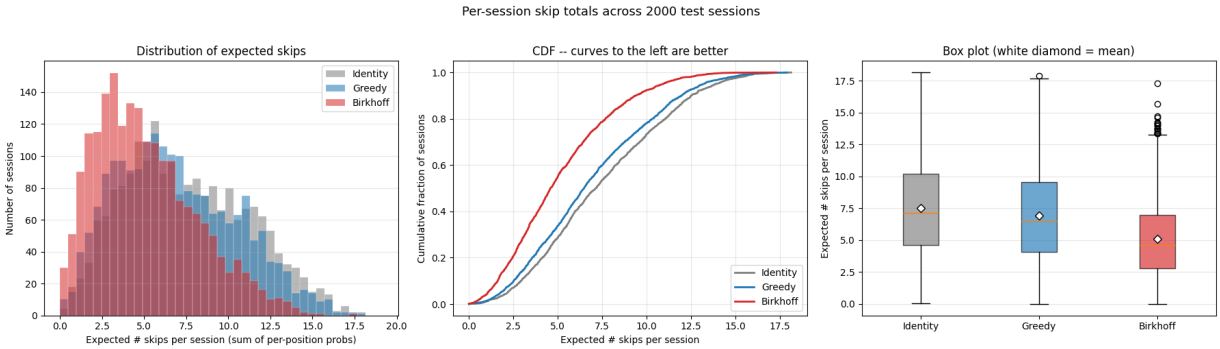

Figure 3: Distribution of expected skips per session across 2,000 held-out sessions. Left: overlaid histograms. Middle: empirical CDFs (curves to the left are better). Right: box plots with the mean marked by a white diamond. Under every summary statistic the Birkhoff distribution is favourable, and the CDF dominates both baselines across the full range.

(white diamond) are all substantially lower than either baseline, while the upper tail is both shorter and bounded by the identity tail.

Figure 4 (next page) plots the Birkhoff skip total against identity (left) and greedy (right) for every test session, with each point coloured by session length n. Points below the dashed line $y = x$ are sessions on which Birkhoff wins. Two patterns are visible. First, almost no sessions sit meaningfully above the diagonal, i.e. Birkhoff rarely makes things worse. Second, the vertical offset below the diagonal grows with the identity skip total, meaning Birkhoff helps most on the sessions where ordering matters most.

Figure 5 (page 10) visualizes what the scorer actually predicts at each position for three illustrative sessions: a large-reduction case, a typical-reduction case, and a pathological case where Birkhoff does not help. In the large-reduction case the session has a high true skip rate (0.85); identity skip probabilities sit near 1.0 at almost every position, while Birkhoff suppresses them to near-zero except at a handful of positions where no good placement exists. In the pathological case the scorer itself assigns high skip probability to nearly every track given the history, and no reordering can escape the ceiling.

Birkhoff is the slowest of the three methods at roughly six seconds per session, dominated by the 200 Adam iterations with 30 inner Sinkhorn steps each. Greedy runs in about one second per session on this length range. On short sessions ($n \leq 20$) the $O(n^2)$ scorer calls of greedy are cheap; the balance would reverse on longer playlists, where greedy's quadratic cost grows faster than Birkhoff's fixed Adam budget.

9 Discussion

The Birkhoff–Greedy gap is larger than the Greedy–Identity gap on average. Greedy finds a strong local improvement because it has position-by-position foresight, but it commits to each placement without reconsidering earlier or later ones. For example, a track that looks locally optimal at position 3 may have a much better global ordering. Birkhoff, because it optimizes a soft surrogate over all positions jointly, can trade local losses for global wins. The per-session pairwise plot (Figure 4) shows the consequence as the vertical spread below the diagonal widens as the original session gets harder, which is exactly where global reshuffling should pay off.

The soft forward $\tilde{f}(\mathbf{P})$ evaluates the scorer on convex mixtures of per-song representations, which the scorer never saw during training. It is therefore miscalibrated as an actual skip probability; empirically it is biased low. We emphasize that this is unproblematic for our pipeline because $\tilde{f}$ is used only as a gradient source, and all reported numbers pass through the hard forward. Future work could reduce the mismatch by annealing the temperature T toward zero during optimization (entropy annealing) or by adding a vector-style regularizer that pushes $\mathbf{P}$ toward the vertices of $\mathbb{B}_n$ during training, in the spirit of the convex-hull characterization in Lim and Wright [12].

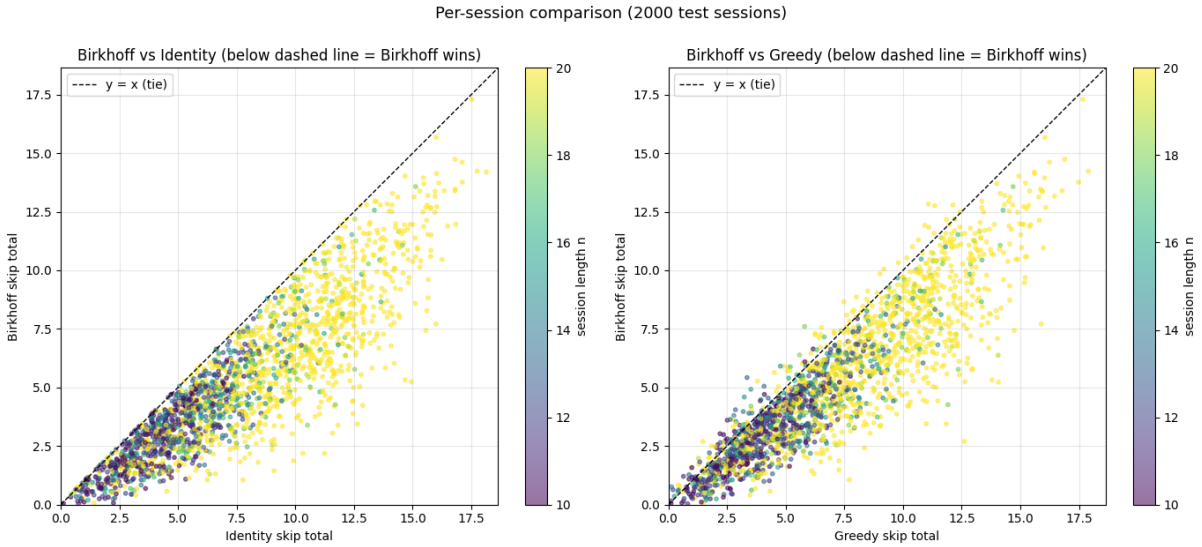

Figure 4: Per-session comparison on all 2,000 test sessions. Birkhoff skip total on the y-axis; identity (left) and greedy (right) on the x-axis. Points are coloured by session length n. Points below the $y = x$ dashed line are sessions where Birkhoff wins. Birkhoff beats identity on 97.7% of sessions and matches or beats greedy on 93.0%; the gains scale with the difficulty of the original session.

Our decoupling is deliberate as it is not a theoretical necessity. One could alternate between scorer training and ordering optimization in an expectation–maximization style. We opted against this for the reasons in Section 6.1: the clean gradient semantics, the honest evaluation, and the match to a deployment scenario in which a fixed production scorer is used to choose orderings at serving time. Joint training is a natural next experiment but requires a careful guard against the trivial minimizer.

Every number in this paper is a *predicted* skip total under a frozen learned scorer. A real listener is not obligated to follow the scorer's beliefs about their behavior, and as such there is no unambiguous ground truth. The honest read of our result is that an ordering policy exists that the best available model of listener skip behavior strongly prefers over identity. Whether a similar ordering would reduce real-world skips is an A/B question for a live service and depends on how well f generalizes.

There are three main limitations worth pointing out. *(i)* We evaluate only on sessions of length 10–20; Birkhoff's wall-clock advantage over greedy should grow with n but we have not characterized the crossover empirically. *(ii)* Position 0 does not contribute to the objective because the scorer requires a non-empty history; we leave the first track fixed to the original. *(iii)* Our scorer is trained on a heavily filtered public dataset that is relatively small; feature distributions in production logs may differ and could require scorer recalibration before the ordering policy is useful.

10 Conclusion and Future Work

We presented a two-module approach to playlist reordering: a context-fused transformer that predicts per-track skip probability, and a Birkhoff-polytope relaxation that searches the space of orderings by gradient descent over doubly-stochastic matrices. The scorer is trained once and frozen; the optimizer is applied at inference time to individual sessions; evaluation is always by a hard, integer-indexed forward pass through the frozen scorer. On 2,000 held-out Spotify sessions the pipeline reduces expected skips by 32.98% on average relative to the original ordering, beating a quadratic greedy search on 93.0% of sessions.

Several directions are natural extensions. *Multi-objective ordering.* Product teams may care about artist fairness, intra-session diversity, or novelty in addition to skip rate; these are easy to incorporate as additional soft terms in $\tilde{f}$ without changing the Birkhoff machinery. *Longer playlists.* The Adam-on-$\mathbf{L}$ budget is fixed in n, while greedy scales as $O(n^2)$ scorer calls and enumeration as $O(n!)$; measuring the wall-clock crossover at $n \gtrsim 50$ is a

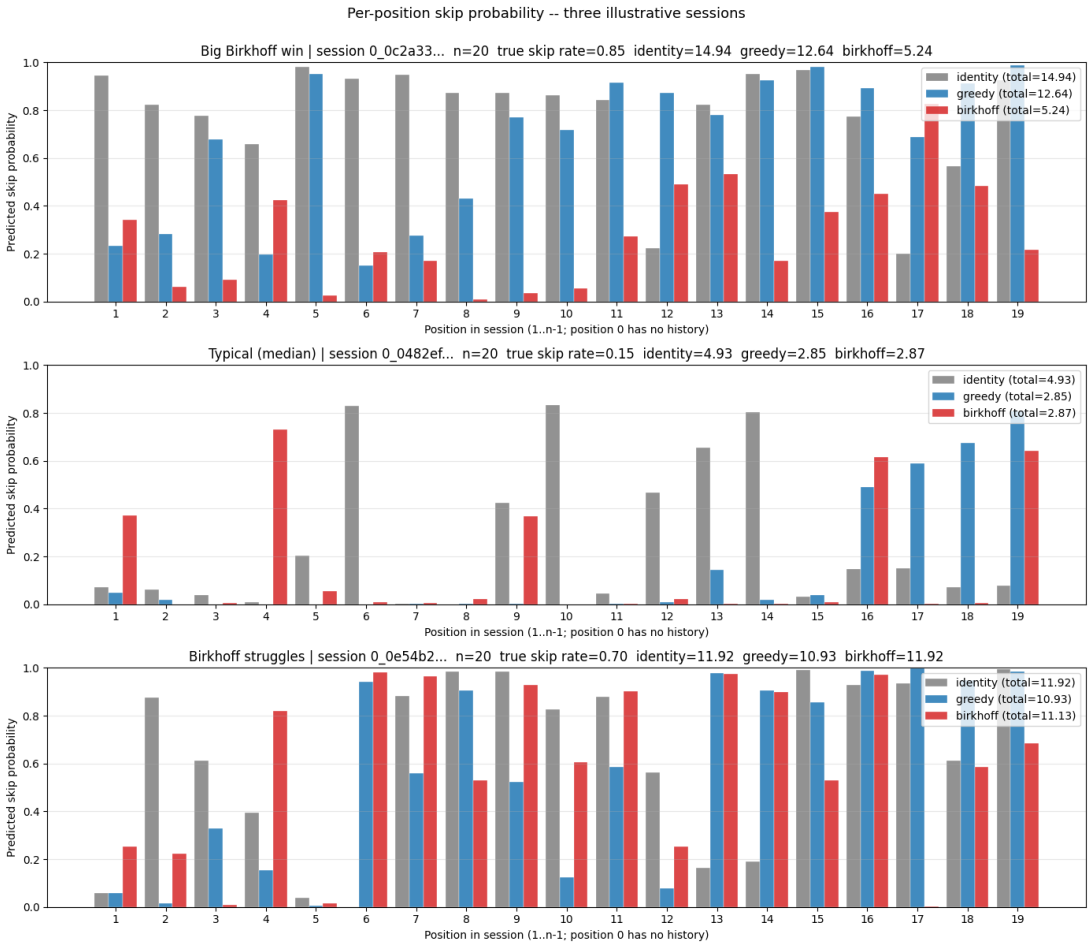

Figure 5: Per-position predicted skip probabilities for three representative sessions. Top: a session where Birkhoff reduces the total from 14.94 to 5.24. Middle: a typical-reduction session. Bottom: a session where the scorer itself predicts high skip probability at most positions, so no reordering can help. Position 0 is omitted because it carries no history under the objective.

good experiment to try. *Cross-session personalization.* Our scorer uses within-session history only; a user-level memory could be added to f without changing the optimizer at all. *Production-grade A/B.* The next step is a live experiment measuring observed skip rates under the Birkhoff policy.

References

[1] Ryan Prescott Adams and Richard S. Zemel. Ranking via Sinkhorn propagation. arXiv:1106.1925, 2011.

[2] Garrett Birkhoff. Tres observaciones sobre el álgebra lineal. *Revista Universidad Nacional de Tucumán, Serie A*, 5: 147–151, 1946.

[3] Brian Brost, Rishabh Mehrotra, and Tristan Jehan. The music streaming sessions dataset. In *Proceedings of The Web Conference (WWW)*, 2019.

[4] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2013.

[5] Aditya Grover, Eric Wang, Aaron Zweig, and Stefano Ermon. Stochastic optimization of sorting networks via continuous relaxations. In *International Conference on Learning Representations (ICLR)*, 2019.

[6] Balázs Hidasi, Alexandros Karatzoglou, Linas Baltrunas, and Domonkos Tikk. Session-based recommendations with recurrent neural networks. In *International Conference on Learning Representations (ICLR)*, 2016.

[7] Eric Jang, Shixiang Gu, and Ben Poole. Categorical reparameterization with Gumbel-Softmax. In *International Conference on Learning Representations (ICLR)*, 2017.

[8] Roy Jonker and Anton Volgenant. A shortest augmenting path algorithm for dense and sparse linear assignment problems. *Computing*, 38(4):325–340, 1987.

[9] Wang-Cheng Kang and Julian McAuley. Self-attentive sequential recommendation. In *IEEE International Conference on Data Mining (ICDM)*, 2018.

[10] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.

[11] Harold W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2): 83–97, 1955.

[12] Cong Han Lim and Stephen J. Wright. Beyond the Birkhoff polytope: Convex relaxations for vector permutation problems. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014. arXiv:1407.6609.

[13] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[14] Chris J. Maddison, Andriy Mnih, and Yee Whye Teh. The concrete distribution: A continuous relaxation of discrete random variables. In *International Conference on Learning Representations (ICLR)*, 2017.

[15] Gonzalo Mena, David Belanger, Scott Linderman, and Jasper Snoek. Learning latent permutations with Gumbel-Sinkhorn networks. In *International Conference on Learning Representations (ICLR)*, 2018.

[16] Rodrigo Santa Cruz, Basura Fernando, Anoop Cherian, and Stephen Gould. DeepPermNet: Visual permutation learning. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[17] Noam Shazeer. GLU variants improve transformer. arXiv:2002.05202, 2020.

[18] Richard Sinkhorn. A relationship between arbitrary positive matrices and doubly stochastic matrices. *The Annals of Mathematical Statistics*, 35(2):876–879, 1964.

[19] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[20] Biao Zhang and Rico Sennrich. Root mean square layer normalization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.